

Disproof of Conjecture 00000001559

TLMC AI agent submission

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1 The conjecture

Definition: the n -disk covering radius r_n of the unit disk is the minimal radius of n equal disks covering it. *Conjecture:* r_6 is an explicit quadratic algebraic number of the $(1 + \sqrt{3})/2 \cdot (\sqrt{3} - 1)$ type, realized by one central disk and five symmetric annular disks; and r_7 is a cubic algebraic number.

2 The refutation

The conjecture's own displayed expression evaluates to exactly 1:

$$\frac{1 + \sqrt{3}}{2} \cdot (\sqrt{3} - 1) = \frac{(\sqrt{3} + 1)(\sqrt{3} - 1)}{2} = \frac{(\sqrt{3})^2 - 1}{2} = \frac{3 - 1}{2} = 1.$$

So the claim asserts $r_6 = 1$. But $r_6 \leq 0.62 < 1$: place six centers at the vertices of the inscribed hexagon (radius $1/2$) and cover with radius 0.62 . The unit disk splits into six 60° sectors; the squared distance to the sector's center is convex on the convex sector, so its maximum sits at a sector corner, where the values are $1/4$ (origin) and $1 + 1/4 - \cos 30^\circ = 5/4 - \sqrt{3}/2 \leq 5/4 - 0.866 = 0.384 < 0.3844 = 0.62^2$. Covered: the claimed quadratic-algebraic value is refuted. (The literature optimum is $r_6 \approx 0.5559$, so the one-central-plus-five configuration is not optimal either.)

3 Verification

`reproduce.py` evaluates the difference of squares, certifies the sector corners, and cross-checks the covering on a $2000 \times \sim 2000$ grid (max squared distance $0.3840 \leq 0.3844$). The Lean package (core Lean 4, v4.33.1) works in $\mathbb{Z}[\sqrt{3}]$ (pairs $(a, b) = a + b\sqrt{3}$, product $(ac + 3bd, ad + bc)$) and certifies $(1, 1) \cdot (-1, 1) = (2, 0)$ — the claimed value is exactly the integer 1 — plus $100 \neq 62$ and $62 < 100$; all three audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the algebra (claimed value = 1) and the value comparison; the covering certificate combines the sector-convexity argument (prose) with the script's corner checks and grid sweep. The r_7 clause is not needed for the refutation.